HDOC DAY – 63

Algorithm

1. Start.
2. Read the size of the array.
3. Declare the variable-length array nums[size].
4. Read all the elements of nums[].
5. Pass the address of nums[] and size to missing_number().
6. Calculate the expected sum of integers from 0 to size using size * (size + 1) / 2.
7. Initialize actual_sum to 0.
8. Traverse the array using a pointer and add each element to actual_sum.
9. Calculate missing = expected_sum - actual_sum.
10. Display the missing number.
11. Stop.

C PROGRAM:

```
#include <stdio.h>


void missing_number(int *nums, int size)
{
   int i;
   int expected_sum = 0;
   int actual_sum = 0;
   int missing;


   expected_sum = size * (size + 1) / 2;


   for (i = 0; i < size; i++)
   {
```

```
        actual_sum = actual_sum + *(nums + i);
    }

    missing = expected_sum - actual_sum;

    printf("Missing number: %d\n", missing);
}

int main()
{
    int size, i;

    printf("Enter the size of the array: ");
    scanf("%d", &size);

    int nums[size];

    printf("Enter the array elements: ");
    for (i = 0; i < size; i++)
    {
        scanf("%d", &nums[i]);
    }

    missing_number(nums, size);

    return 0;
}
```

TEST CASE TABLE:

Test Case	Size	Array	Expected Output
1	3	3 0 1	Missing number: 2
2	2	0 1	Missing number: 2
3	9	9 6 4 2 3 5 7 0 1	Missing number: 8
4	1	0	Missing number: 1
5	5	5 2 1 0 4	Missing number: 3